tion: Division:

e:

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SEMESTER :- 8^{th}

Course CODE :- STAT - 415

n:

PAPER :- Total Quality Management

PAST PAPERS (Theoretical PORTION)

-: 2016 - 2024 :-

issues:

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PAST PAPER 2016

a International Standard Organization

ISO is the international standard Organization It was setup in 1947 and is located in Switzerland. It's purpose to facilate and support national and international trad and commerce by developing standards that people everywhere would recognize and respect.

Benifits of ISO:-

i- Suitable for both large and small organization

2) Better internal managment 3) less wastage

4) Increase in effeciency, productivity and profit.

5) Consistent outcomes, measured and monitored.

b) Attribute Control Charts:-

A control chart for attributes is used to monitor characteristics that have discrete values and can be counted. Often they can be evaluated with the simple yes or no decision. We may judge each unit product as either conforming or non-conforming on the basis

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a) Total Quality Management:-

It is a application of quantitative method's and human resources to improve all the

Distribution:

process within an organization and customer needs now and in the future. TQM integrates fundamental managment techniques, existing improvement efforts and technical

PAR:

tools under a desciplined approach

Competative Bench Marking:-

Competative Bench Marking in Total Quality Managment is a process of

Any other issues:

Comparing an organization's products, services or process with those of its competitions or industry leaders. The goal is to identify best practices, measure Performance gaps, and implement impoement to gain a competitive advantage

xample:-Coca-Cola benchmarking it's supply chain managment agaist Pepsi to enhance efficiency and reduce costs

Signature:

of whether or not it possesses certain attribute:-
<u>Example</u>:- i) The Number of broken cookies in the box
2) No of persons having eye colour "blue" e.t.c

c) Failure Cost in TQM:-

Nonconformance costs are associated with products or services that do not conform the customer's requirement. They are often referred to be as a failure cost. These costs can be catagorized into two main types. Internal Failure Costs:- These costs occure before the product or service is dilived to the customer, such as scrap, Rework.

External Failure costs:- These costs occure after the product or service has been delived to the customer, Such as returns or replacement, Loss of customer loyalty

d) Principal of Quality Managment:-

The principales of Quality managment in TQM include:-
i- Customer Focus:- ii leadership 3 Employee involvement
4 Process Approach 5 Continuous Improrment 6 Data-Driven Decision making 7 Supplier parternship.

iii- ISO:- 2016

iv Failure Cost in TQM 2016

Past Paper 2018:-

1- Dimensions of Quality:-

1) Performance 2) Features 3) Conformable 4) Service
5) Realibility 6) Duribility 7) Respond 8) Asthetic
9) Repulations:-

2) Eras of Quality Managment:-

In 1924, W. A Shewart of Bell Telephone laboratiries developed a statistical chart for a control of product variables, This chart is considered to be the begening of statistical quality control

In 1946, the American society for Quality was formed This organization through it's publications, conferences and training sessions has promote the use of quality control for all type of production and service.

In 1950 W. Edwards Deming who learned statistical quality from Shewhart,

Joseph M. Juran Made his first trip to Japan in 1954 and further emphasized management's resposibility to achieve quality.

In the late 1980s the automotive industry began to emphasize statistical process control (SPC). Suppliers and their suppliers were required to use these techniques.

By the year 2000, the quality focus was shifting to information technology within an organization and externally via the internet.

5 Enhance customer satisfaction:-
Ensure products and services meet customer requirements and expectations.

iv- **External Failure Costs in TQM:-**
External Failure Costs in Total Quality Managment refers to costs associated with defects or failures that occur after a product or service has been delived to the customer. These costs are incude warranty claims, product recalls, customer complaints, loss of customer layality and Damage to reputation. External failure costs can be significant and can harm a campanay's repulation and bottom line. By focusing on quality and prevention, organizations can reduce the liklihood of external failure costs.

Past Paper:- 2019

i **Six Sigma Methodology:-**
Six Sigma follows two main approaches:-
DMAIC (Define, measure, Analyse, improve, Control)
DMADV:- (Define, measure, Analyze, Design, verify).

ii- **Philosophy of TQM:-**
Dr. N Edward Deming (1980-1993) is considered to be the father of modern quality. He preached that to achieve the highest level of performance requires more than a good philosophy the organization must change its behaviour and adopt new ways of doing business:-

iii **Acceptance Sampling plans:-**
Acceptance Sampling is the process of evaluating a portion of the product in a lot for the purpose of accepting or rejecting the entire lot as either conforming or not conforming to a quality specification. ~~Acceptance Sampling is the process of~~ evaluating can be performed by using either attributes measurments or variable measurements.

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Objective of ISO:-

The main objectives of ISO (International Organization for Standaxidazation) are.

i- Promote global Standarization:-

Solution:

Develope and publish international standards for products, services, and processes.

ii- Ensure quality and sefety:-

Provide frameworks for organization to demonstrate quality, safety and efficiency.

iii- Facilitate international trade:-

Harmonize standards across countries to reduce

Issues:

trade barriers and facilities global commerce.

iv- Improve efficiency and productivity:- Help organizations streamline processes, reduce waste, and improve overall performance

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Internal Failure Costs in TQM:-

Internal failure costs are a costs associated with defects found before the customer receives the product or Service.

ion:

PAST PAPER 2020:-

i- Process Capability

process capability in quality Control managment refers to the ability of the process to produce output within specified limits or tolalances. It

ssues:

measures how well a process can meet Customer requirements or Specifications.

Signature:

• Eras of quality management:-

2018

• Rectifying Inspection:-

Rectifying insepection is a type of incpection where defective products or items are identified, removed, and replaced or reworked to meet quality standards. The goal is ensure that only products that meet specifications are delivered to customers.

It involves:-

1- Identying defects
2- Removing defects
3- Verifying quality

• External Failure Costs in TQM:-

2018

Past Paper:- 2022

i- Statistical Quality Control:-

Statistical Quality Control is a methodology that uses statistical methods to monitor, control and improve the quality of process and products. It involves collecting and analyzing data to identify variations, detect defects and make informed decisions to ensure quality standards are met.

SQC techniques includes:-

i Control Charts
ii Statistical Process Control (SPC)
iii Accepting Sampling
iv Design of experiment (DOE)

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b:- Warning limits :-

Warning limits are set closer to the mean ~~that~~ than action limits - usually at $\pm 2\sigma$. Crossing these limits is a sign of potential instability, but not necessarily an out of contool process

Distribution:

Purpose:-

- Signal that the process may be trending

PAR:

- Sugguest increased monitoring
- Help to detecting problems early, before action limits are closed

c:- Natural torelance limit :-

Any other issues:

Natural torelance limits, also Known as 3-Sigma limits, are boundries that define the inherent variabilitY of a process based on it's standard deviation. They represent the expected range of values with in which a process should operate while maintaining statistical control, typically encompassing 99.37% of the data.

Signature:

d Average Outgoing quality:-

Average Outgoing quality is a quality control matric that Calculate the expected quality of products or services that leave a production process after inspection. It helps access the effectiveness of quality control measures and identifies areas for improvement. AOQ is calculated based on the quality of incoming products and effectiveness of the inspection process, especially the rectification of rejected lots.

e Process Capability :-

2020

-: Producer's and Consumer's Risk :-

Producer's Risk

Defination:-
The probability of rejecting a good-quality batch of products

Consequences:-
If the jet manufacturer accepts a defective engine and the the complete jet cannot be sold, the jet with a defective engine is sold to an airline the airline could experience a financial loss and other business problems, such as a loss in repulalation.

Causes:-

Consumer's Risk.

Defination:-
The probability of accepting a defective Product as good.

Consequences:-
If the jet manufacturer doesn't accept the engines the firm creates, for example the firm could experience financial strain because a jet engine is a highly-price commodity

Causes:-

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Past Paper:- 2023

a) Action Limits :-

Action limits are control chart boundries that, when exceeded, indicate a high probability of process instability or a defect. These limits typically correspond to ± 3 standard deviation (σ) from the process mean.

Distribution:

PAR:

Purpose :-

- Signal that the process is out of control.
- Require immediate corrective action.
- Indicate assignable cause variation (not random chance).

ny other issues:

Signature:

Example :-

If the mean of a process is 100 with a standard deviation of 2.

Action limits = $100 \pm 3 \times 2 \rightarrow 94$ to 106

b:- Objectives of ISO :-

2018

c:- Natural tolerance limits :-

2022

d:- Acceptance Sampling :-

2019

e:- Total Quality Managment

2017

f:- Bench marking Process.

is a process measuring and comparing an organizations' processes, products or services against industry bests or best practices from other industries to identify areas for improvement.

Reasons for bench marking:

i performance improvement:- Helps organizations to ~~help~~ identify gaps and improve efficiency.

ii Competitive Advantage:- Enables businesses to stay competive by adopting best practices

iii Customer Satisfaction:- Enhance Quality, leading to better customer experiences.

iv Cost Reduction :- identifies ineffeciencies and helps in reducing operational costs.

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Past Paper :- 2024

i- Total Quality Managment 2017.

ii Specification limits :-

Distribution:

The boundries that define the acceptable range of values for a quality characteristic or process They are essentially the "acceptable" values for a product or process.

PAR:

set by customer requirements, industry standards, or internal engineering specifications:

Any other issues:

3 Natural tolerance limit's 2022

4 Average Out going Quality

4 Advantages of Double Sampling Plan :-

Signature:

The advantages of a double sampling plane includes.

i:- Reduce Sampling Cost :- Double sampling plans can be more cost-effective than single sampling plans, as they often require small initial sample sizes.

ii :- Improve decision making :- double sampling plans ~~can b~~ allows for a second sample to be taken before making a final decision, which can be reduce the risk of incorrect decision

iii Increased accuracy :- By taking a second sample double sampling plans can provide more accurate estimates of the population quality.

iv Flexibility :-

Double sampling plans can be designed to accomodate different sampling risks and quality levels

v Six Sigma Methodology :-

2019

vi Rectifying Inspection :-

2022.